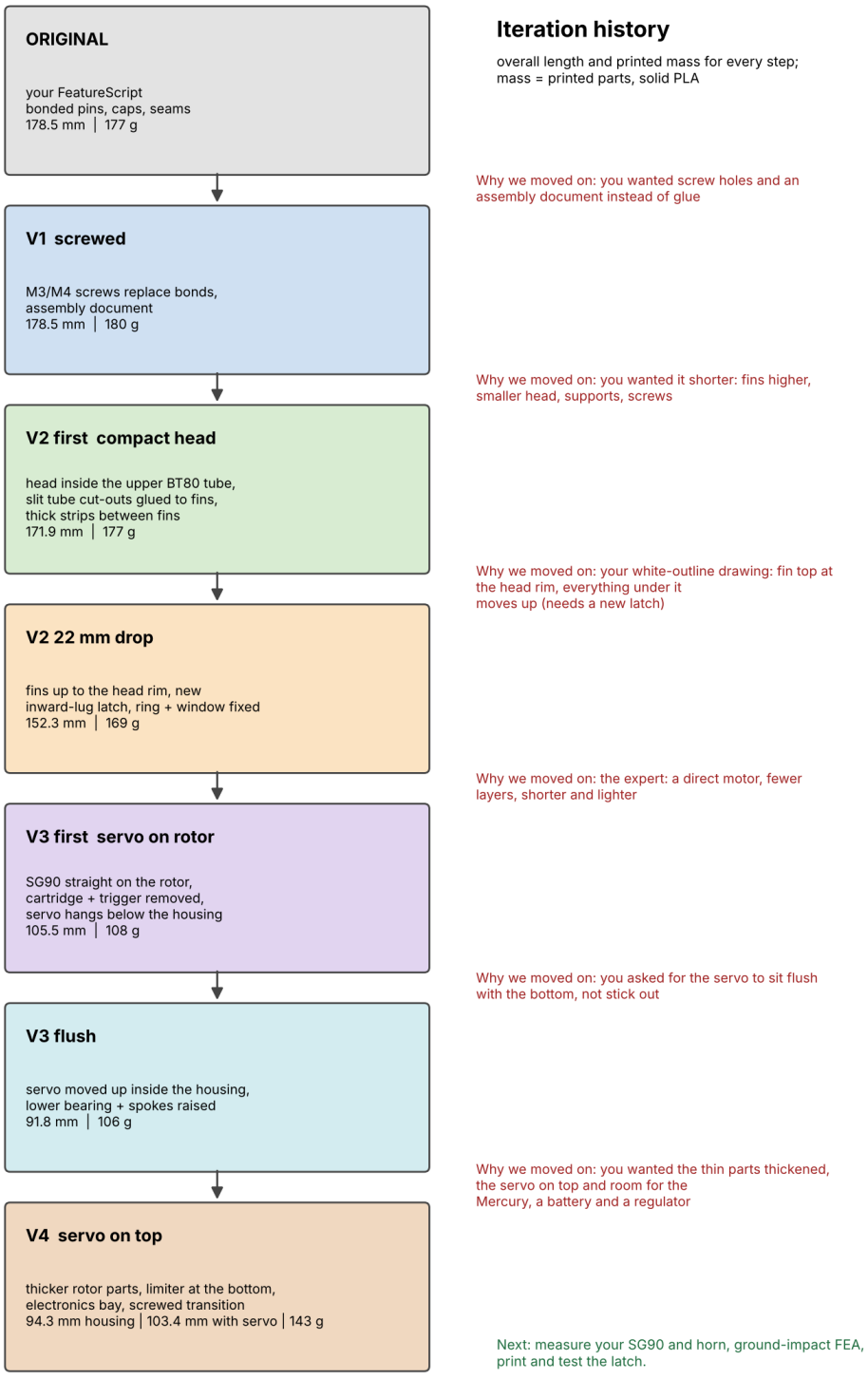
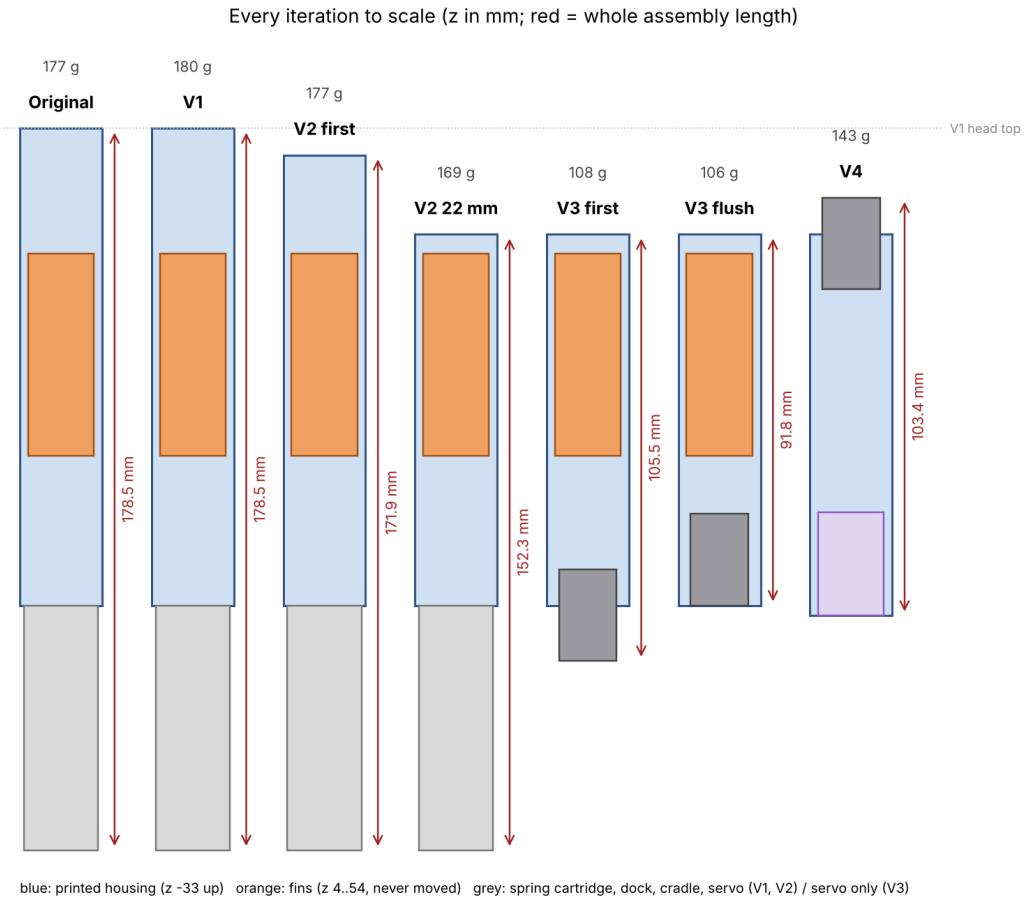
# Design history: every iteration and why we switched

From your original FeatureScript to the servo-on-top V4 with its electronics bay. Numbers are measured on the CAD models (printed mass = solid PLA at 1.24 g/cm³; the original and the first V2 were measured from their saved models and STL exports).

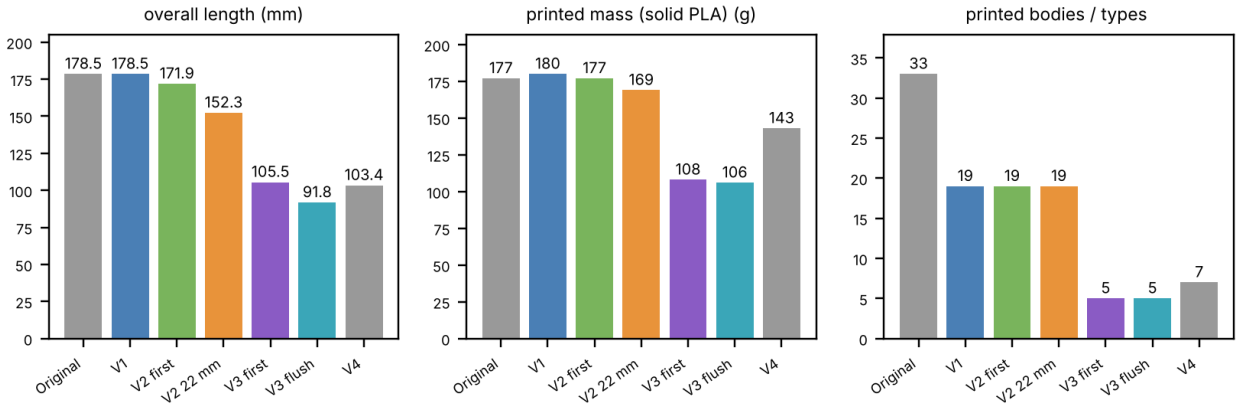



The path: seven steps, each triggered by a request or a finding.

# 1 Side by side




To scale. The fins never moved (z 4..54). What changed is what sits above and below them.

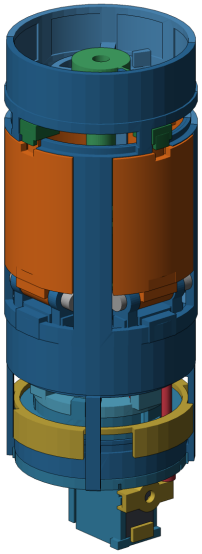



Length, mass and number of printed parts for each iteration.

| | Printed housing | Overall length | Printed mass | Printed parts | Drive and latch |
|---|---|---|---|---|---|
| Original | 118 mm | 178.5 mm | 177 g | 33 bodies (glued pins, caps) | wound spring + SG90 trigger; tab-and-hook latch on the rotor |
| V1 screwed | 118 mm | 178.5 mm | 180 g | 19 types | same |
| V2 first (compact head) | 111.4 mm | 171.9 mm | 177 g | 19 types | same |
| V2 22 mm drop | 91.8 mm | 152.3 mm | 169 g | 19 types | spring + SG90 trigger; inward lug + finger latch |
| V3 first (servo below) | 91.8 mm | 105.5 mm incl. servo | 108 g | 5 types | SG90 on the rotor; same lug + finger latch |
| V3 flush (servo inside) | 91.8 mm | 91.8 mm | 106 g | 5 types | same; bearing and spokes raised |

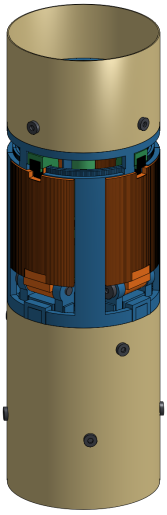
| | Printed housing | Overall length | Printed mass | Printed parts | Drive and latch |
|---|---|---|---|---|---|
| V4 (servo on top, bay) | 96.3 mm | 105.4 mm (servo top above the housing) | 144 g incl. sled + transition | 7 types | servo on top, thick limiter at the bottom |

## 2 The original: your FeatureScript



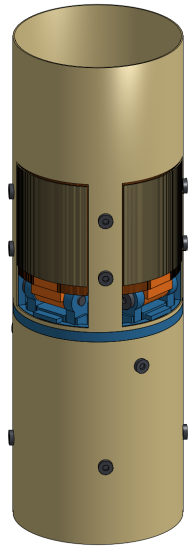
- Source: your “Positive cam kick-out / modular brake” FeatureScript, with your unloaded-spring print references; I rebuilt it in a new Onshape document and matched its size (z –93.5…85) and part count before changing anything.
- How it works: a wound spiral spring in a cartridge drives a four-lobe cam rotor; an SG90 pulls a trigger through a pushrod and yoke; a hook on the rotor holds a tab on each fin; the cams kick the fins and the airflow finishes opening them.
- Fastening: glue. Hinge pins H and caps K, the catch caps, the housing seam, the glue sleeves, the lid.
- **178.5 mm overall, 177 g, 33 printed bodies.**
- **Why we moved on:** you asked for screw holes in place of the bonds, and an assembly document.

## 3 V1: the same machine, screwed together

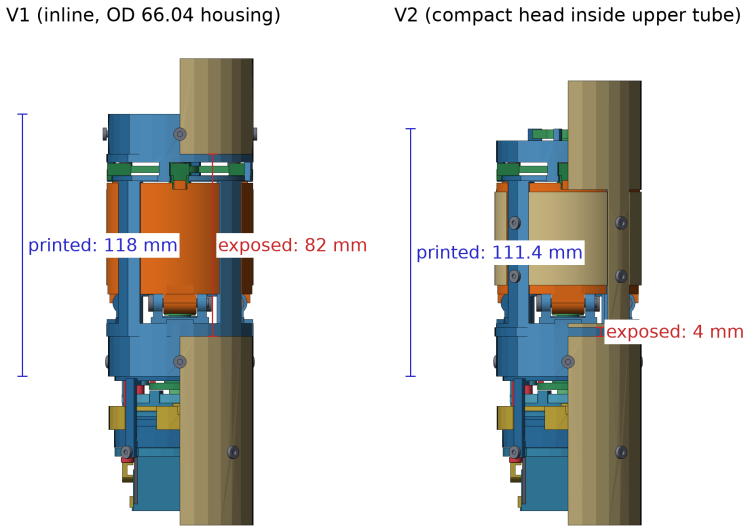


- Seam: 4 × M3×8 socket-head screws across the housing seam on inward ribs at z = 8 and z = 44. Hinges: M4×25 screw per fin instead of pin and cap. Catch caps: open-top clip with an M3×8 into the rotor arm. Sleeves, dock ring, cradle and lid: M3×6 and M3×10 radial or vertical screws.
- Stayed bonded: the thrust collar, servo keeper and return-leaf root (under 3 mm of material, a screw would be weaker than the glue).
- Checked: no overlaps, straight hex-key access to the seam screws with the fins open.
- **178.5 mm overall, 180 g, 26 screws, 56 bodies incl. hardware.**
- Found and fixed on the way: a casing boss that made a non-manifold boolean; lid ears that hit the pushrod (moved to 145/215/335°); seam screws at z = 6 blocked by hinge-ear tops (moved to z = 8).
- **Why we moved on:** you wanted a shorter airbrake: fins higher, a smaller head (fin thickness adds to the diameter), thick supports, screw holes.

# 4 V2 first: compact head inside the upper BT80 tube



- Your first brief asked for 4 supports between the fins and 4 under them. You then proposed something simpler and better: cut slits in the upper BT80 tube, glue each cut-out to its fin so it rides with the fin, keep supports only in the solid strips between the slits, and keep the sleeve fit at the bottom.
- Fin shell R29.9–32.3 (64.6 mm OD), 0.2 mm glue gap, then the 0.52 mm BT80 wall: back to 66.04 mm flush. The upper sleeve is gone; 8 supports (4 strips, 4 stop blocks) take 8 × M3 into the tube.
- Cam peak 29 → 28.6 mm for pillar clearance. Tube windows 38.6 mm wide; cut-outs start at z = 8 so they clear the stop block at 90°; tube holes centred on the strips (offset ones left only 1.3 mm of web).
- **Exposed airbrake 82 → 4 mm. Housing 111.4 mm, overall 171.9 mm, 177 g.**
- **Why we moved on:** your white-outline drawing on a screenshot: the fins should sit up at the rim of the head, so everything under the fin moves up and the whole piece is shorter than V1.



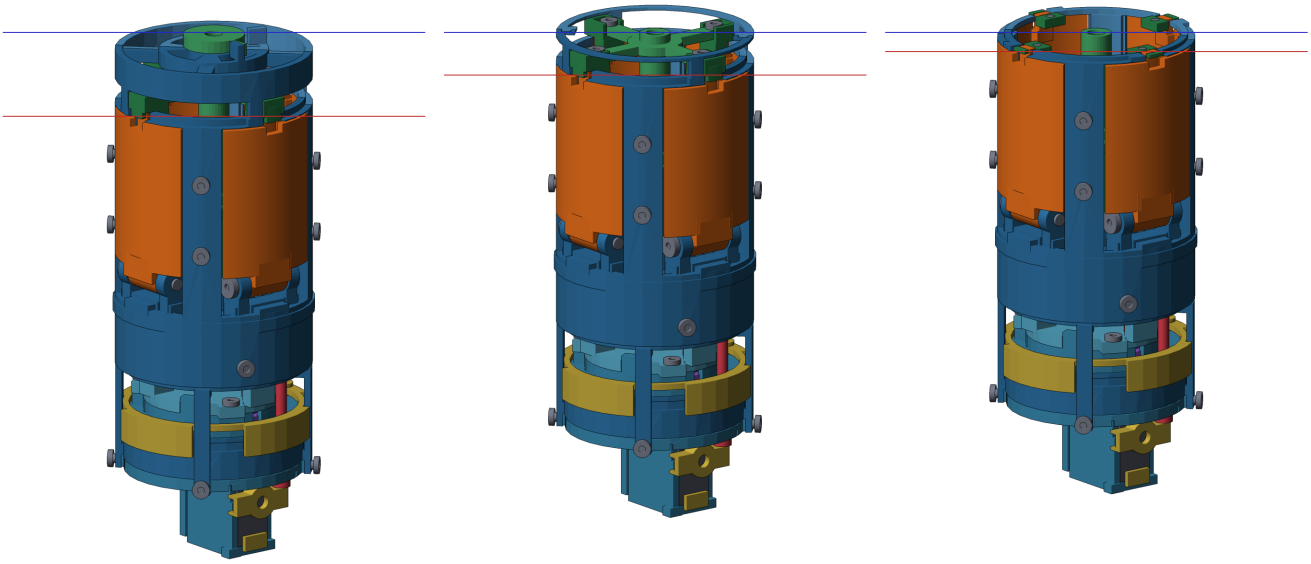

V1 (left) and the first V2 (right), as built at the time.

# 5 V2 at 22 mm: the latch stack moves up into the fin height

V2 now
printed stack 171.9 mm (V1: 178.5)
fin top to head top: 28.4 mm

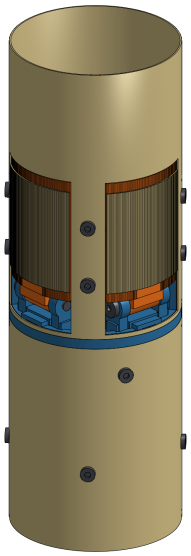
latch stack down 14 mm
printed stack 157.9 mm (V1: 178.5)
fin top to head top: 14.4 mm

latch stack down 22 mm
printed stack 149.9 mm (V1: 178.5)
fin top to head top: 6.4 mm



Cut-away preview of the real V2 model: red = fin top, blue = head top. Latch stack not yet re-engineered (see text).

The preview I made so you could choose: the real V2 with the top trimmed by 0, 14 and 22 mm. You chose 22 mm. (The mm figures in the picture were the estimate; the final build is 152.3 mm overall.)



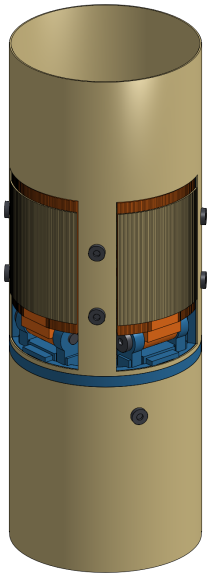
- Problem: the V1 hook swings sideways through the fin's own window, so it cannot fit inside the fin height. New latch: each fin carries a lug that reaches inward (plate + wall), and the catch cap on the rotor arm carries a finger just outside that wall. Measured on the CAD: held from 2° of opening, released by 16° of rotor travel, before the cams touch at about 26°.
- Everything below the fin (rotor stack, bearings, knob, caps) moved up 22 mm. Fin top z 54, head top z 58.8 (4.8 mm).
- **91.8 mm housing, 152.3 mm overall, 169 g.**
- Corrections after the first 22 mm build: the window was 60.4 mm tall and the top ring overlapped the fin top by 231 mm³ (window now 63.8 mm, ring raised: 89.4 → 91.8 mm); a 0.1 mm skin in the catch-cap socket; and, found later, the reference tube still had a 60.2 mm window, which the fin tip would have hit (now 63.8 mm; swept 0.5–90°, clear).
- **Why we moved on:** an experienced person you talked to: make it a direct motor drive and drop the layers; shorter and lighter.

## 6 V3 first: the SG90 drives the rotor directly

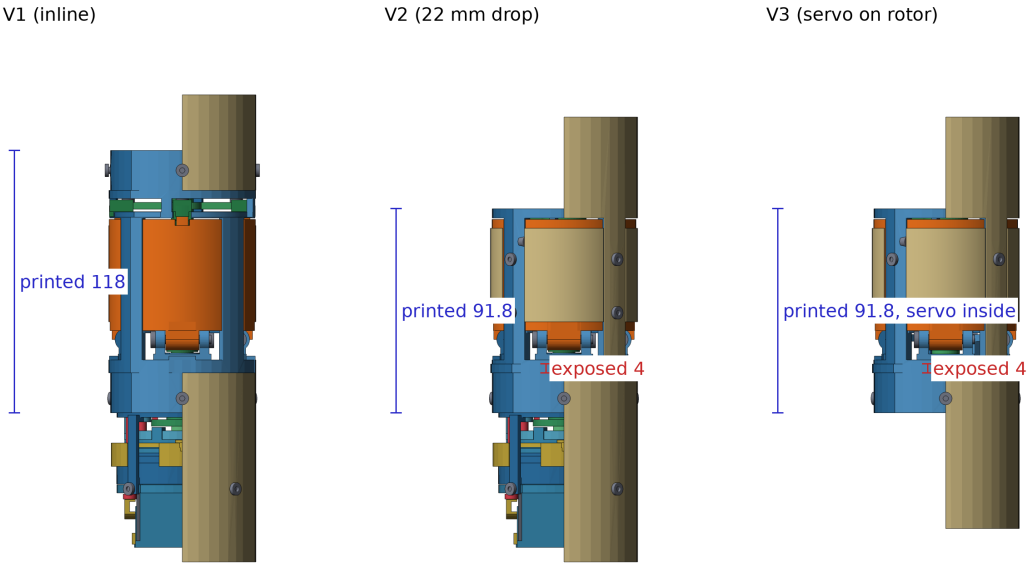
Removed: spring cartridge, trigger, pushrod, yoke, dock, casing, lid, hub, collar, pin, cradle, keeper, horn saddle (about 64 g printed), plus the spring and return leaf. Added: a horn socket in the rotor underside, two posts that hold the SG90 by its flange, three pads for the lower tube. Latch, cams, fins and tube slits are the V2 ones, so the same checks apply.

- Torque (SG90): the load that matters is friction at the latch while the wind pulls the closed fins outward; roughly fine up to 77–109 m/s in a pessimistic case, 134–188 m/s in the typical one.
- Deploy once and stay out: the 90° hard stop is in the fin geometry; the cam only holds the fins at about 14°, the airflow does the rest.
- **105.5 mm overall, 108 g, 5 printed part types.** The servo hung about 13.7 mm below the bottom of the housing.
- **Why we moved on:** you asked for the servo to be moved up so it sits flush with the bottom and does not stick out, while still working.

## 7 V3 flush: the servo inside the housing

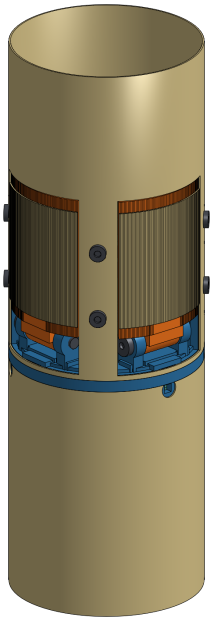


- Moving the servo up alone was not possible: the rotor's lower bearing and its four spokes sat exactly where the servo has to go (z –15…–6).
- So the lower bearing (now 3 mm, z –0.8…2.2), the thrust collar and the spokes moved **up** above the horn, the rotor now ends at z –8.7, the horn pocket sits at z –8.5…–6.7, and the servo posts hang from the raised spokes. The servo's bottom is exactly at the housing bottom (z –33).
- A 0.7 mm relief in the base ring clears the servo tab. Latch, cam timing and the fin motion are unchanged (checked again on the new model).
- Checked: no printed-part overlaps closed or open; rotor, horn and caps swept 0–65° clear of housing and servo; hex-key access to the seam screws clear; ASM-V3 mates (21) all solve.
- **91.8 mm overall, 106 g, 5 printed part types** (V2: 152.3 mm and 169 g).
- **Next:** measure your own servo and horn, run the ground-impact FEA you asked about, print and test the latch.

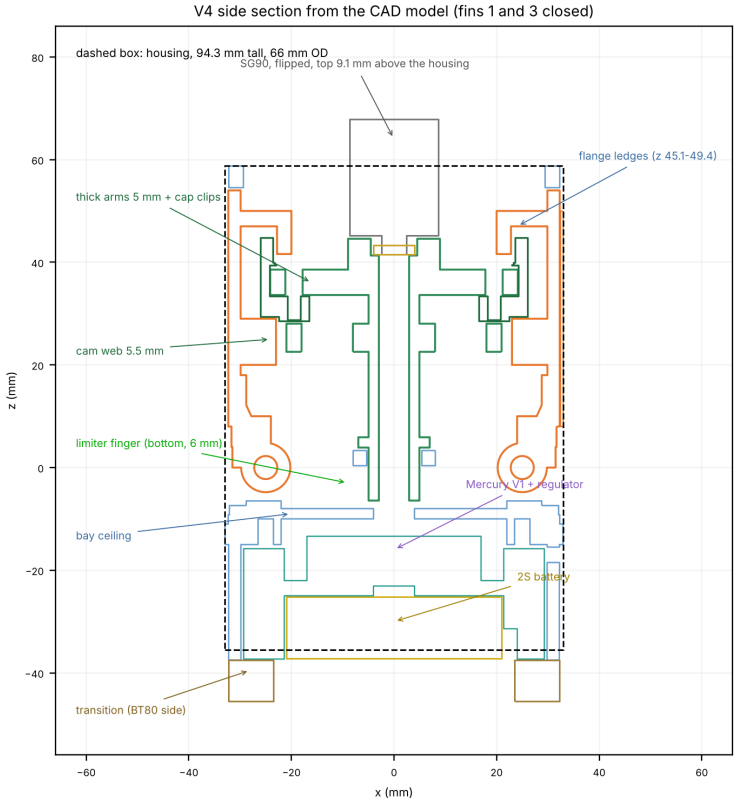



V1, V2 and V3 side by side (V3 is the flush layout).

# 8 V4: servo on top, thicker parts, electronics bay



- You noticed the limiter stops and the green rotor parts were thin, and asked for the servo on top with the limiter moved to the bottom, a screwed bottom joint on the BT80 transition, and a bay for the altimeter, a servo battery and a regulator, keeping the size small.
- I drew a sketch first (no CAD) and you picked Option B (a little more length), the Mercury V1 altimeter, and my battery assumption.
- Thicker: limiter finger 2 → 6 mm, stops 2 → 7.6 mm, crown arms 3 × 5 → 5 × 8 mm (walls at the screw 0.8 → 2.3 mm), cam web 3 → 5.5 mm.
- Bay: sled with the Mercury and a regulator on one layer and the battery on the other; four vertical M3 into a transition ring.
- The interference sweep showed the servo ledges were hit by the catch-cap fingers, so the ledges moved above them and the servo now stands 9.1 mm above the housing.
- **Housing 96.3 mm, 105.4 mm from housing bottom to servo top, printed set 144 g** (114 g core + sled + transition ring).




V4 section from the CAD model.

## 9 Problems found along the way and what fixed them

| Step | Problem | Fix |
|---|---|---|
| Setup | Free Onshape plan forces public documents; your own document was read-only | New public document (you chose to keep it public) |
| Setup | Direct STL export was blocked by the proxy | STLs built from the model's tessellation; STEP through Onshape's translation |
| V1 | Casing boss tangent to the cylinder gave a non-manifold boolean | Boss embedded into the casing (r0 = 22.0) |
| V1 | Lid ears hit the pushrod; one boss 0.1 mm too tall | Ears at 145/215/335°, boss height 4.0 mm |
| V1 | Seam screws at z = 6 blocked by hinge-ear tops | Moved to z = 8 (hex-key ray test now clear) |
| V2 first | Tube cut-outs clipped the stop block at 90°; offset tube holes left a 1.3 mm web | Cut-outs start at z = 8; holes centred on the strips |
| V2 22 mm | V1 hook cannot fit inside the fin height | Inward lug on the fin + finger on the catch cap |
| V2 22 mm | Fin window too short: ring overlapped the fin top by 231 mm³ | Window 63.8 mm, ring raised (housing 91.8 mm) |
| V2 22 mm | Reference tube window still 60.2 mm; fin tip would hit it while opening | Window 63.8 mm; opening sweep 0.5–90° clear |
| V2 22 mm | My angle scan polled Onshape too hard and was blocked for about 20 h | Waited; later checks done offline on the meshes (no API calls) |
| V3 flush | Servo cannot just move up: rotor bearing and spokes were in the way; tab 0.5 mm into the base ring | Bearing, collar and spokes raised above the horn; 0.7 mm ring relief |
| V4 | Catch-cap fingers hit the servo flange ledges between 30° and 60° of rotor travel (63 mm³) | Ledges raised above the fingers; servo stands 9.1 mm above the housing |
| V4 | Limiter finger only touched the shaft (stayed a separate body); seam pegs floated after the base ring was opened; sled overlapped the hinge ears | Finger overlapped into the shaft, pegs moved to R28.5, sled ribs trimmed to the lower bay |
| All | Assembly mate values cannot be driven from a script | Not verified by script; drag the rotor and a fin in ASM-V3 / ASM-V4 |

## 10 Who decided what

| Decision | By |
|---|---|
| Screws instead of glue, M3 self-tapping, M4 for the hinge, public Onshape document | You |
| Shorter airbrake with the fins higher and a smaller head | You |
| Slit the upper tube and glue the cut-outs to the fins; supports only in the strips; sleeve fit at the bottom | You (replaced my first 8-support plan) |
| Fin top at the rim of the head; 22 mm drop rather than 14 mm | You |
| New lug-and-finger latch, 63.8 mm window, 4.2 mm top ring | Me (needed to make your layout work) |
| Direct SG90 drive, deploy once, 90° mechanical stop, FEA later | You (from the expert's advice) |
| Servo flush with the housing bottom, not sticking out | You |
| Servo on top, limiter to the bottom, thicker parts, screwed transition, electronics bay, Option B, Mercury V1 | You |
| Servo posts and pads, horn pocket, cam peak 28.6 mm, small joints left bonded | Me; please check the servo and horn dimensions |
| Moving the lower bearing and spokes up to make the flush servo fit | Me |
| Raising the servo ledges above the finger sweep; sled, transition ring, regulator placeholder | Me |
| Battery strap channels in the sled and notches in the transition ring | You asked; me |
| Battery pocket resized for the Spektrum 300 mAh 2S (42 × 19 × 12 mm) | You asked; me |

Mass is solid PLA; the slicer's infill will lower the real weights. The latch, horn pocket and servo mount have not been tested on a print.